# ST5202 — Applied Regression Analysis Lecture Notes 2. Multiple Linear Regression Models

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# Outline

§1. Multiple LSM and its Estimation

§2. Inference with LSE

# §1.1. Multiple linear regression model (LRM)

- In a multiple linear regression analysis, we deal with the relationship between a response variable $Y$ and $p$ predictor variables $\boldsymbol{X} = (X_1, ..., X_p)$. The relation is investigated through a data set of $n$ observations:

| obs. | $Y$ | $X_1$ | $X_2$ | ... | $X_p$ |
|---|---|---|---|---|---|
| 1 | $y_1$ | $x_{11}$ | $x_{12}$ | ... | $x_{1p}$ |
| 2 | $y_2$ | $x_{21}$ | $x_{22}$ | ... | $x_{2p}$ |
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ |
| n | $y_n$ | $x_{n1}$ | $x_{n2}$ | ... | $x_{np}$ |

- The multiple linear regression model between $Y$ and $\boldsymbol{X}$ is given by

$$Y = \beta_0 + \beta_1 X_1 + ... + \beta_p X_p + \epsilon,$$

where $\epsilon$ is a random error with mean zero, and $\beta_0, \beta_1, ..., \beta_p$ are unknown regression parameters. The function

$$Y(\boldsymbol{X}) = \beta_0 + \beta_1 X_1 + ... + \beta_p X_p$$

is called the multiple linear regression function.

# §1.2. Multiple LRM for observed data

A multiple LRM with $n$ observations is given by

$$\begin{aligned} y_1 &= \beta_0 + \beta_1 x_{11} + \ldots + \beta_p x_{1p} + \epsilon_1, \\ y_2 &= \beta_0 + \beta_1 x_{21} + \ldots + \beta_p x_{2p} + \epsilon_2, \\ &\vdots \\ y_n &= \beta_0 + \beta_1 x_{n1} + \ldots + \beta_p x_{np} + \epsilon_n. \end{aligned}$$

In short,

$$y_i = \beta_0 + \beta_1 x_{i1} + \ldots + \beta_p x_{ip} + \epsilon_i, \quad i = 1, \ldots, n.$$

# §1.3. The matrix form of multiple LRM

Let

$$X = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1,p} \\ 1 & x_{21} & \cdots & x_{2,p} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{n,p} \end{bmatrix}, \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}, \boldsymbol{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \boldsymbol{\epsilon} = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}.$$

We then have

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} \beta_0 + \beta_1 x_{1,1} + \ldots + \beta_p x_{1,p} + \epsilon_1 \\ \beta_0 + \beta_1 x_{2,1} + \ldots + \beta_p x_{2,p} + \epsilon_2 \\ \vdots \\ \beta_0 + \beta_1 x_{n,1} + \ldots + \beta_p x_{n,p} + \epsilon_p \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1,p} \\ 1 & x_{21} & \cdots & x_{2,p} \\ \vdots & \vdots & & \vdots \\ 1 & X_{n1} & \cdots & X_{n,p} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}.$$

In short,

$$\boldsymbol{y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}.$$

$X$ is referred to as the design matrix.

# §1.4. The assumptions for multiple LRM

The assumptions are the same as those for simple LRM.

(i) The $X_1, \ldots, X_p$ and $\epsilon$ are independent.

(ii) The $\epsilon_i$'s have mean zero.

(iii) The $\epsilon_i$'s are pairwise un-correlated, i.e., $\mathrm{Cov}(\epsilon_i, \epsilon_j) = 0$.

(iv) The $\epsilon_i$'s have common variance $\sigma^2$ (Homogeneity).

(v) The $\epsilon_i$'s have a Normal distribution (Normality).

Let $\mu_y$ denote the mean of $Y$, $\mu_x$ the mean vector of $X = (X_1, \ldots, X_p)^\top$, $\Sigma_{xy}$ the covariance vector between $Y$ and $X$, $\Sigma_{xx}$ the covariance matrix of $X$, and $\boldsymbol{\beta}_1 = (\beta_1, \ldots, \beta_p)^\top$. By assumption (i), taking Cov and E on both sides of $y = X^\top \boldsymbol{\beta} + \epsilon$ and solving, we have

$$\boldsymbol{\beta}_1 = \Sigma_{xx}^{-1} \Sigma_{xy}, \ \beta_0 = \mu_y - \boldsymbol{\beta}_1^\top \boldsymbol{\mu}_x.$$

# §1.5. The least square estimation for multiple LRM

- The least square estimation estimates the parameters $\beta_0, \beta_1, \ldots, \beta_p$ by minimizing

$$\begin{aligned} Q &= \sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_{i1} - \cdots - \beta_p x_{ip})^2 = \|\boldsymbol{y} - X\boldsymbol{\beta}\|_2^2 \\ &= (\boldsymbol{y} - X\boldsymbol{\beta})^\top(\boldsymbol{y} - X\boldsymbol{\beta}) = \boldsymbol{y}^\top\boldsymbol{y} - 2\boldsymbol{y}^\top X\boldsymbol{\beta} + \boldsymbol{\beta}^\top X^\top X\boldsymbol{\beta}. \end{aligned}$$

- By using the matrix calculus,

$$\frac{\partial Q}{\partial \boldsymbol{\beta}} = -2X^\top\boldsymbol{y} + 2X^\top X\boldsymbol{\beta}.$$

  Setting the derivative to zero, we get the normal equation:

$$X^\top X\boldsymbol{\beta} = X^\top\boldsymbol{y}.$$

  The LSE of $\boldsymbol{\beta}$ is then given by $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1}X^\top\boldsymbol{y}$.

- The $\sigma^2$ is estimated by

$$\hat{\sigma}^2 = \frac{\|\boldsymbol{y} - X\hat{\boldsymbol{\beta}}\|^2}{n-p-1} = \frac{\|\boldsymbol{y} - X(X^\top X)^{-1}X^\top\boldsymbol{y}\|^2}{n-p-1} = \frac{\|\boldsymbol{y} - H\boldsymbol{y}\|^2}{n-p-1}.$$

# §1.6. The explicit expression of $\hat{\beta}_j$

- The explicit expression of $\hat{\beta}_j$ can be obtained by considering the minimization of $\|\boldsymbol{y} - X\boldsymbol{\beta}\|_2^2$ in two steps: first minimize w.r.t. $\boldsymbol{\beta}_{-j}$ with $\beta_j$ fixed, then minimize w.r.t. $\beta_j$, where $\boldsymbol{\beta}_{-j}$ is the sub-vector of $\boldsymbol{\beta}$ eliminating $\beta_j$, that is, $\min_{\beta_j} \min_{\boldsymbol{\beta}_{-j}} \|\boldsymbol{y} - X\boldsymbol{\beta}\|_2^2$. We have

$$\begin{aligned} \min_{\beta_j} \min_{\boldsymbol{\beta}_{-j}} \|\boldsymbol{y} - X\boldsymbol{\beta}\|_2^2 &= \min_{\beta_j} \min_{\boldsymbol{\beta}_{-j}} \|\boldsymbol{y} - \beta_j \boldsymbol{x}_j - X(-j)\boldsymbol{\beta}_{-j}\|_2^2 \\ &= \min_{\beta_j} \|(I - H_{-j})(\boldsymbol{y} - \beta_j \boldsymbol{x}_j)\|_2^2 \\ &= \min_{\beta_j} \|[I - H_{-j}]\boldsymbol{y} - \beta_j [I - H_{-j}]\boldsymbol{x}_j\|_2^2. \end{aligned}$$

- The minimizer of $\beta_j$ is then

$$\hat{\beta}_j = \frac{\boldsymbol{x}_j^\top (I - H_{-j})\boldsymbol{y}}{\boldsymbol{x}_j^\top (I - H_{-j})\boldsymbol{x}_j}.$$

- Let $\tilde{\boldsymbol{y}} = (I - H_{-j})\boldsymbol{y}$, $\tilde{\boldsymbol{x}}_j = (I - H_{-j})\boldsymbol{x}_j$. $\hat{\beta}_j$ is the estimate of $\tilde{\beta}$ in

$$\tilde{y} = \tilde{\beta}\tilde{x}_j + \tilde{\epsilon}.$$

# §1.7. The hat matrix and its properties

- The matrix $H = X(X^\top X)^{-1}X^\top$ is called the hat matrix of $X$. It is the projection matrix of the linear space spanned by the columns of $X$.
- The hat matrix has the properties:
  - $H^2 = H,\ (I-H)^2 = I - H$ .
  - $HX = X$, which implies

$$H\mathbf{1} = \mathbf{1},\ H\boldsymbol{x}_j = \boldsymbol{x}_j,\ j = 1, \ldots, p,$$

  where $\mathbf{1} = (1, \ldots, 1)^\top$, $\boldsymbol{x}_j = (x_{1j}, \ldots, x_{nj})^\top$.
- The vector of fitted values and residual vector are expressed in terms of $H$ as,

$$\hat{\boldsymbol{y}} = H\boldsymbol{y},\ \boldsymbol{e} = \boldsymbol{y} - \hat{\boldsymbol{y}} = (I - H)\boldsymbol{y},$$

  Immediately, we have

$$\boldsymbol{e}^\top\mathbf{1} = 0,\ \boldsymbol{e}^\top\boldsymbol{x}_j = 0,\ \bar{\hat{y}} = \bar{y}.$$

# §1.8. Sum of Squares in matrix form

As in the simple LRM case, the total variation in $Y$ is $\text{SST} = \sum_{i=1}^{n}(y_i - \bar{y})^2$, the variation due to regression is $\text{SSR} = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2$, and the variation due to random error is $\text{SSE} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$.

- For SST, consider the vector

$$\boldsymbol{y}_T = \begin{pmatrix} y_1 - \bar{y} \\ \cdots \\ y_n - \bar{y} \end{pmatrix} = \boldsymbol{y} - \mathbf{1}\bar{y} = (I - \frac{\mathbf{1}\mathbf{1}^\top}{n})\boldsymbol{y} \equiv H_T\boldsymbol{y}.$$

  Thus, $\text{SST} = \boldsymbol{y}_T^\top \boldsymbol{y}_T = \boldsymbol{y}^\top H_T^\top H_T \boldsymbol{y}$.

- For SSR, consider the vector

$$\boldsymbol{y}_R = \begin{pmatrix} \hat{y}_1 - \bar{y} \\ \cdots \\ \hat{y}_n - \bar{y} \end{pmatrix} = \hat{\boldsymbol{y}} - \mathbf{1}\bar{y} = [X(X^\top X)^{-1}X^\top - \frac{\mathbf{1}\mathbf{1}^\top}{n}]\boldsymbol{y} \equiv H_R\boldsymbol{y}.$$

  Thus, $\text{SSR} = \boldsymbol{y}_R^\top \boldsymbol{y}_R = \boldsymbol{y}^\top H_R^\top H_R \boldsymbol{y}$.

- For SSE, consider

$$\boldsymbol{e} = \boldsymbol{y} - \hat{\boldsymbol{y}} = (I - X(X^\top X)^{-1}X^\top)\boldsymbol{y} \equiv H_E\boldsymbol{y}.$$

  Thus, $\text{SSE} = \boldsymbol{e}^\top \boldsymbol{e} = \boldsymbol{y}^\top H_E^\top H_E \boldsymbol{y}$.

# §1.9. Decomposition of Sum of Squares

- It can be easily verified that the matrices $H_T, H_R$ and $H_E$ are all symmetric and idempotent. Hence,

$$\begin{aligned}
\text{SST} &= \boldsymbol{y}^\top H_T \boldsymbol{y} = \boldsymbol{y}^\top (I - \frac{\mathbf{1}\mathbf{1}^\top}{n})\boldsymbol{y}, \\
\text{SSR} &= \boldsymbol{y}^\top H_R \boldsymbol{y} = \boldsymbol{y}^\top [X(X^\top X)^{-1}X^\top - \frac{\mathbf{1}\mathbf{1}^\top}{n}]\boldsymbol{y}, \\
\text{SSE} &= \boldsymbol{y}^\top H_E \boldsymbol{y} = \boldsymbol{y}^\top [I - X(X^\top X)^{-1}X^\top]\boldsymbol{y}.
\end{aligned}$$

- Using the matrix forms, we have

$$\begin{aligned}
\text{SST} &= \boldsymbol{y}^\top (I - \frac{\mathbf{1}\mathbf{1}^\top}{n})\boldsymbol{y} \\
&= \boldsymbol{y}^\top [X(X^\top X)^{-1}X^\top - \frac{\mathbf{1}\mathbf{1}^\top}{n} + I - X(X^\top X)^{-1}X^\top]\boldsymbol{y} \\
&= \text{SSR} + \text{SSE}.
\end{aligned}$$

# §1.10. Degrees of freedom of the sum of squares

- SST, SSR and SSE are in fact the squared norms of the $n$-vectors $\boldsymbol{y}_T, \boldsymbol{y}_R$ and $\boldsymbol{e}$ respectively.
- Since $\boldsymbol{y}_T = H_T\boldsymbol{y}, \boldsymbol{y}_R = H_R\boldsymbol{y}$, $\boldsymbol{e} = H_E\boldsymbol{y}$,
$$\boldsymbol{y}_T \in \mathcal{R}(H_T), \quad \boldsymbol{y}_R \in \mathcal{R}(H_R), \quad \boldsymbol{e} \in \mathcal{R}(H_E),$$
where $\mathcal{R}(A)$ denotes the subspace of $R^n$ spanned by the columns of $A$.
- $H_T$, $H_R$ and $H_E$ have ranks $n-1$, $p$ and $n-p-1$ respectively. The ranks are the dimensions of the subspaces spanned by these matrices.
- For a subspace $\mathcal{R}_1$ of $R^n$ with dimension $k$, there is a subspace $\mathcal{R}_2$ with dimension $n-k$ which is orthogonal to $\mathcal{R}_1$. Hence, there are $n-k$ independent vectors in $\mathcal{R}_2$, say, $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{n-k}$, which are perpendicular to the vectors in $\mathcal{R}_1$. Thus, for any vector $\boldsymbol{u}$ in $\mathcal{R}_1$, $\boldsymbol{u}^\top \boldsymbol{v}_j = 0$, $j = 1, \ldots, n-k$, that is, only $k$ components of $\boldsymbol{u}$ can vary freely. So we say that the degrees of freedom (d.f.) of $\boldsymbol{u}$ is $k$. Note that the d.f. is the same as the dimension of $\mathcal{R}_1$.
- The d.f. of the norm of a vector refers to the d.f. of the vector. Thus the d.f. of SST, SSR and SSE are $n-1$, $p$ and $n-p-1$ respectively.

# §1.11. ANOVA Tables

- Ordinary ANOVA table:

| Source | DF | SS | MS | F-statistic |
|---|---|---|---|---|
| Regression | $p$ | $SSR$ | $MSR = SSR/p$ | MSR/MSE |
| Error | $n-p-1$ | $SSE$ | $MSE = \frac{SSE}{n-p-1}$ | |
| Total | $n-1$ | $SST$ | | |

- Sequential ANOVA table:

| Source | DF | SS | MS | F-statistic |
|---|---|---|---|---|
| $X_1$ | 1 | $SS_1$ | $MS_1 = SS_1$ | $MS_1/MSE$ |
| $X_2$ | 1 | $SS_2$ | $MS_2 = SS_2$ | $MS_2/MSE$ |
| $\cdots$ | $\cdots$ | $\cdots$ | $\cdots$ | $\cdots$ |
| $X_p$ | 1 | $SS_p$ | $MS_p = SS_p$ | $MS_p/MSE$ |
| Error | $n-p-1$ | $SSE$ | $MSE = \frac{SSE}{n-p-1}$ | |
| Total | $n-1$ | $SST$ | | |

# §1.12 Sequential sum of squares

- The sum of squares in the sequential ANOVA table, which is generated by the R function `anova`, are sequential sum of squares. The sequential sum of squares associated with $X_j$ is the SSR of a simple LRM involving $X_j$ while the effect of $X_j$ is adjusted for the effects of all the covariates preceding $X_j$.
- Specifically, $SS_1$ is the SSR of the model

$$\boldsymbol{y} = \beta_0 + \beta \boldsymbol{x}_1 + \boldsymbol{\epsilon},$$

  $SS_j$ is the SSR of the model

$$\tilde{\boldsymbol{y}} = \tilde{\beta} \tilde{\boldsymbol{x}}_j + \tilde{\boldsymbol{\epsilon}},$$

  where $\tilde{\boldsymbol{y}} = (I - H_{j-1})\boldsymbol{y}$, $\tilde{\boldsymbol{x}}_j = (I - H_{j-1})\boldsymbol{x}_j$ and $H_{j-1}$ is the hat matrix of $(\mathbf{1}, \boldsymbol{x}_1, \ldots, \boldsymbol{x}_{j-1})$, $\tilde{\boldsymbol{x}}_j = (I - H_{j-1})\boldsymbol{x}_j$.
- The SSR of the multiple regression model is the sum of the sequential sums of squares.
- The sequential ANOVA table cannot be used to make inference on each of the individual covariates except the last one.

# §1.13. The coefficient of multiple determination

- The same as for simple LRM, the coefficient of multiple determination for multiple LRM, which measures the proportion of the variation in the response variable explained by the predictor variables, is defined as
$$R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = \frac{\text{SSR}}{\text{SST}}.$$
- $R^2$ can also be expressed as $R^2 = \text{corr}(\boldsymbol{y}, \hat{\boldsymbol{y}})^2$, where
$$\text{corr}(\boldsymbol{y}, \hat{\boldsymbol{y}}) = \frac{\sum_{i=1}^n (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{[\sum_{i=1}^n (y_i - \bar{y})^2 \sum_{i=1}^n (\hat{y}_i - \bar{\hat{y}})^2]^{1/2}}.$$
For the verification, note that
$$\begin{aligned}\sum_{i=1}^n (\hat{y}_i - \bar{y})^2 &= \sum_{i=1}^n (\hat{y}_i - \bar{\hat{y}})^2 = \boldsymbol{y}^\top [X(X^\top X)^{-1} X^\top - \frac{\mathbf{1}\mathbf{1}^\top}{n}]\boldsymbol{y} \\ &= \boldsymbol{y}^\top [X(X^\top X)^{-1} X^\top - \frac{\mathbf{1}\mathbf{1}^\top}{n}](I - \frac{\mathbf{1}\mathbf{1}^\top}{n})\boldsymbol{y} \\ &= (\hat{\boldsymbol{y}} - \bar{\hat{\boldsymbol{y}}})^\top (\boldsymbol{y} - \bar{\boldsymbol{y}}) = \sum_{i=1}^n (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}}).\end{aligned}$$
- The adjusted $R^2$: $\quad R_a^2 = R^2 - \frac{p}{n-p-1}(1 - R^2)$.

# §1.14. Multiple correlation coefficient (1)

- The correlation of the response variable $Y$ with a scalar covariate is measured by the Pearson's correlation coefficient. How to measure the correlation between the response variable with a vector $\boldsymbol{x}$ of covariates?
- We can consider any linear combination of $\boldsymbol{x}$, say, $\boldsymbol{a}^\top \boldsymbol{x}$, and consider the Pearson's correlation between $Y$ and $\boldsymbol{a}^\top \boldsymbol{x}$. The largest Pearson's correlation among all possible $\boldsymbol{a}$ can be used to measure the correlation between $Y$ and $\boldsymbol{x}$.
- The multiple correlation coefficient between a random variable $Y$ and a random vector $\boldsymbol{x} = (X_1, \ldots, X_p)^\top$ is defined as

$$\mathrm{MCORR}(Y, \boldsymbol{x}) = \max_{\boldsymbol{a}} \mathrm{CORR}(Y, \boldsymbol{a}^\top \boldsymbol{x}),$$

where $\boldsymbol{a}$ is any vector of non-random variables.

# §1.15. Multiple correlation coefficient (2)

- For the multiple correlation coefficient, we have

$$[\text{MCORR}(Y, \boldsymbol{x})]^2 = \frac{\Sigma_{y\boldsymbol{x}}\Sigma_{\boldsymbol{xx}}^{-1}\Sigma_{\boldsymbol{x}y}}{\sigma_y^2},$$

  where $\Sigma_{y\boldsymbol{x}} = (\text{Cov}(Y, X_1), \cdots, \text{Cov}(Y, X_p))$, $\Sigma_{\boldsymbol{xx}}$ is the variance matrix of $\boldsymbol{x}$ and $\Sigma_{\boldsymbol{x}y} = \Sigma_{y\boldsymbol{x}}^\top$, $\sigma_y^2 = \text{Var}(Y)$.

- Let $\boldsymbol{y}$ be the vector of $n$ observations of $y$, $X$ be the matrix of observed $\boldsymbol{x}$ with its $i$th row given by $\boldsymbol{x}_i^\top$. The sample multiple correlation coefficient $\text{MCORR}(\boldsymbol{y}, X)$ is defined similarly and we have

$$[\text{MCORR}(\boldsymbol{y}, X)]^2 = \frac{\boldsymbol{y}^\top(I - \frac{\mathbf{1}\mathbf{1}^\top}{n})X[X^\top(I - \frac{\mathbf{1}\mathbf{1}^\top}{n})X]^{-1}X^\top(I - \frac{\mathbf{1}\mathbf{1}^\top}{n})\boldsymbol{y}}{\boldsymbol{y}^\top(I - \frac{\mathbf{1}\mathbf{1}^\top}{n})\boldsymbol{y}}.$$

- In multiple LRM with a vector of $p$ covariates $\boldsymbol{x} = (X_1, \ldots, X_p)$, we have

$$R^2 = [\text{MCORR}(\boldsymbol{y}, X)]^2.$$

- The details are given in the supplementary note to Lecture notes 2.

# §1.16. Partial correlation coefficients

- Let $X(-j)$ denote the sub-matrix of the design matrix $X$ obtained by deleting column $\boldsymbol{x}_j$ of $X$. Let $\tilde{\boldsymbol{y}}$ be the residual of $\boldsymbol{y}$ regressed on $X(-j)$, $\tilde{\boldsymbol{x}}_j$ the residual of $\boldsymbol{x}_j$ regressed on $X(-j)$. The correlation between $\tilde{\boldsymbol{y}}$ and $\tilde{\boldsymbol{x}}_j$ is called partial correlation between $\boldsymbol{y}$ and $\boldsymbol{x}_j$ adjusting for the effects of $X(-j)$, which is given by

$$\begin{aligned}
\mathrm{CORR}(\tilde{\boldsymbol{y}}, \tilde{\boldsymbol{x}}_j) &= \frac{\sum_{i=1}^n (\tilde{y}_i - \bar{\tilde{y}})(\tilde{x}_{ij} - \bar{\tilde{x}}_j)}{\sqrt{\sum_{i=1}^n (\tilde{y}_i - \bar{\tilde{y}})^2 \sum_{i=1}^n (\tilde{x}_{ij} - \bar{\tilde{x}}_j)^2}} \\
&= \frac{\sum_{i=1}^n \tilde{y}_i \tilde{x}_{ij}}{\sqrt{\sum_{i=1}^n \tilde{y}_i^2 \sum_{i=1}^n \tilde{x}_{ij}^2}} \\
&= \frac{\boldsymbol{y}^\top (I - H_{-j}) \boldsymbol{x}_j}{\sqrt{\boldsymbol{y}^\top (I - H_{-j}) \boldsymbol{y} \boldsymbol{x}_j^\top (I - H_{-j}) \boldsymbol{x}_j}},
\end{aligned}$$

  where $H_{-j}$ is the hat matrix of $X(-j)$.

- For justification, note that, since $\tilde{\boldsymbol{y}}^\top \mathbf{1} = 0, \tilde{\boldsymbol{x}}_j^\top \mathbf{1} = 0$, we have $\bar{\tilde{y}} = 0, \bar{\tilde{x}}_j = 0$.

# §1.17. Interpretation of partial correlation coefficient

- In simple LRM with response $\boldsymbol{y}$ and covariate $\boldsymbol{x}$,

$$R^2 = \text{corr}^2(\boldsymbol{y}, \boldsymbol{x}).$$

- Let $\tilde{R}^2$ be the coefficient of determination of the regression of $\tilde{\boldsymbol{y}}$ on $\tilde{\boldsymbol{x}}_j$. We also have the relationship

$$\tilde{R}^2 = \text{corr}^2(\tilde{\boldsymbol{y}}, \tilde{\boldsymbol{x}}_j).$$

- Since the residual of $\boldsymbol{y}$ regressed on $X(-j)$ is the part of $\boldsymbol{y}$ unexplained by $X(-j)$, the squared partial correlation between $\boldsymbol{y}$ and $\boldsymbol{x}_j$ adjusting for the effects of $X(-j)$ is the proportion of the unexplained variation of $\boldsymbol{y}$ which is further explained by $\boldsymbol{x}_j$.
- On page §1.6, we have that $\hat{\beta}_j$ is the estimate of $\tilde{\beta}$ in

$$\tilde{y} = \tilde{\beta}\tilde{x}_j + \tilde{\epsilon}.$$

Hence, $\hat{\beta}_j$ measures the effect of $x_j$ on $y$ after adjusting for the effects of all the other predictor variables.

# §1.18. A general procedure for practical modeling

- In practical problems, the relationship between a response variable $Y$ and several covariates $X_1, \ldots, X_p$ is to be investigated. The relationship of $Y$ with a particular $X_j$ is not necessarily linear, instead, $Y$ might be related to a non-linear function of $X_j$. The relationship can be modeled by a multiple LRM with $X_j$ replaced by the non-linear function of $X_j$.
- The first step of modeling is to find out the appropriate form of each covariate to be included in the regression model. This can be done by the scatter plots of the response variable against each of the covariates.
- After fitting the regression model, the model adequacy must be checked by various tools, the details of model checking will be discussed in later lectures. For the time-being, only residual plots will be used.
- When model adequacy is confirmed, then proceed to make inference on the model.

# §1.19. Patient Satisfaction (PS) example: Data

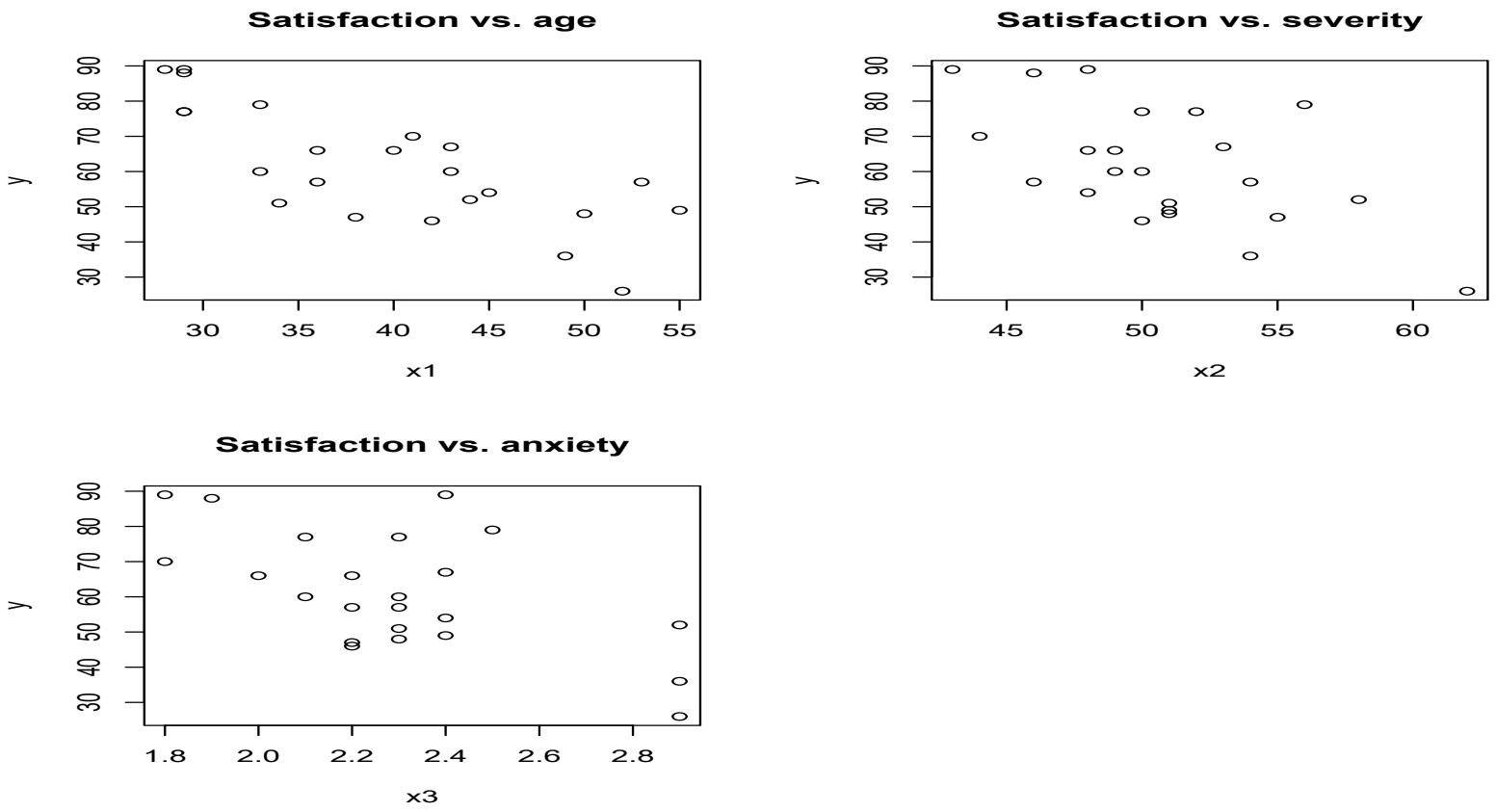
A hospital administration wished to study the relation between patient satisfaction ($Y$) and patient's age ($X_1$, in years), severity of illness ($X_2$, an index), and anxiety level ($X_3$, an index), where larger values of $Y$, $X_2$ and $X_3$ are, respectively, associated with more satisfaction, increased severity of illness and more anxiety. The data for 23 randomly selected patients are in the file `PatientSatisfaction.txt` on Canvas. A portion of the data is as follows:

| p.score | p.age | severity | anxiety |
|---|---|---|---|
| 48.0 | 50.0 | 51.0 | 2.3 |
| 57.0 | 36.0 | 46.0 | 2.3 |
| 66.0 | 40.0 | 48.0 | 2.2 |
| 70.0 | 41.0 | 44.0 | 1.8 |
| 89.0 | 28.0 | 43.0 | 1.8 |

# §1.20. PS example: R code for preliminary modeling

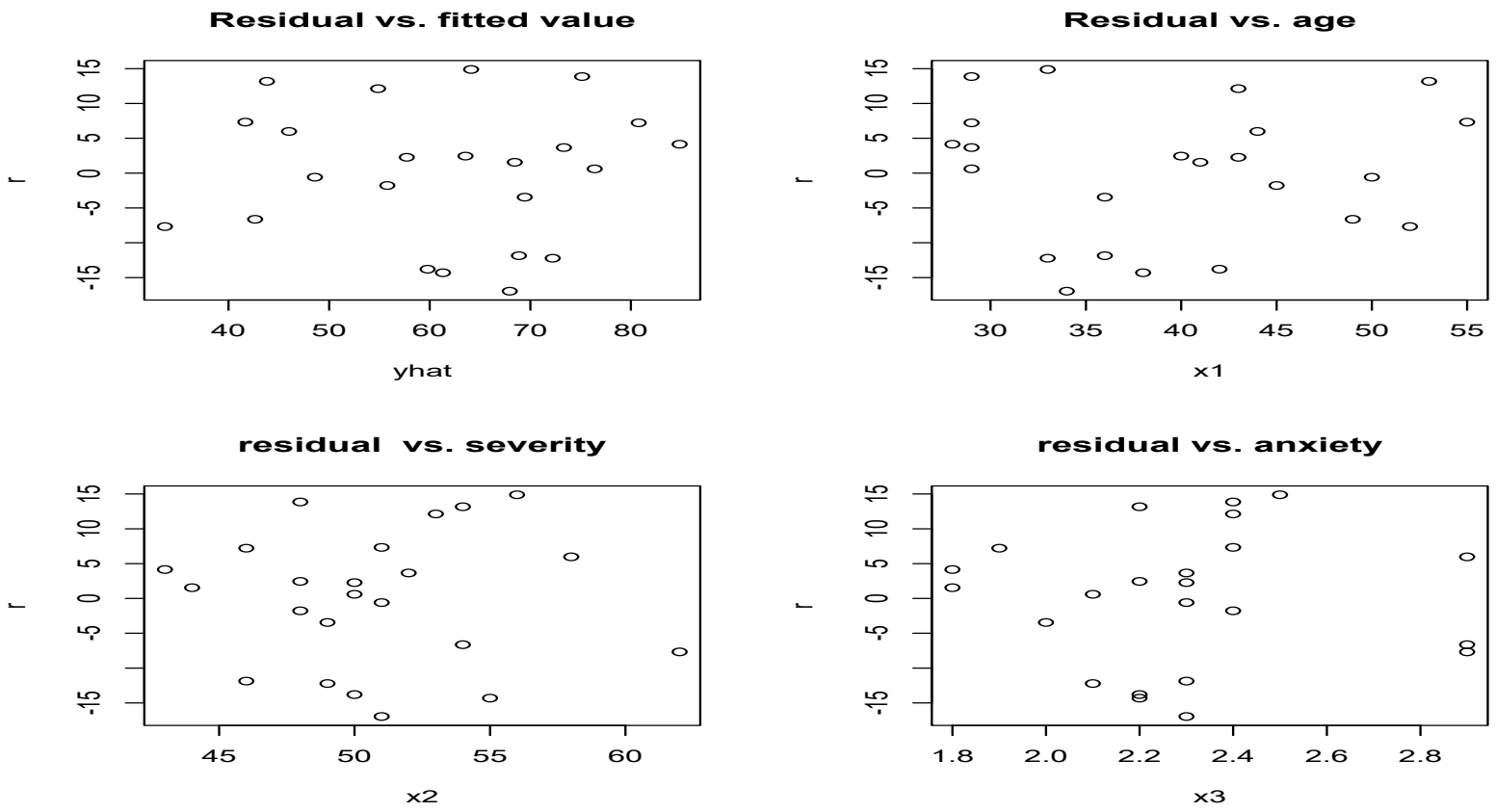
```
PS.dat=read.table("D://Rsession/PatientSatisfaction.txt",header=TRUE)
y=PS.dat$p.score;x1=PS.dat$p.age;x2=PS.dat$severity;x3=PS.dat$anxiety

# scatter plots of y against each covariate
  par(mfrow=c(2,2))
    plot(x1,y,main="Satisfaction vs. age")
    plot(x2,y,main="Satisfaction vs. severity")
    plot(x3,y,main="Satisfaction vs. anxiety")

# Model fitting and checking
  PS.fit = lm(y~x1+x2+x3)
  r = PS.fit$resid
  yhat = PS.fit$fitted
  par(mfrow=c(2,2))
    plot(yhat, r,main="Residual vs. fitted value")
    plot(x1,r,main="Residual vs. age")
    plot(x2,r,main="residual  vs. severity")
    plot(x3,r,main="residual vs. anxiety")
```

# §1.21. PS example: Scatter plots of y against covariates




The plots suggest that the response variable has a linear relation with each of the three predictor variables.

# §1.22. PS example: Residual plots




There is not special pattern showing up in these plots, which suggest the model is appropriate.

# §1.23. PS example: fitted results

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 162.8759    25.7757   6.319 4.59e-06 ***
x1           -1.2103     0.3015  -4.015  0.00074 ***
x2           -0.6659     0.8210  -0.811  0.42736
x3           -8.6130    12.2413  -0.704  0.49021

Residual standard error: 10.29 on 19 degrees of freedom
Multiple R-squared:  0.6727, Adjusted R-squared:  0.621
F-statistic: 13.01 on 3 and 19 DF,  p-value: 7.482e-05

Analysis of Variance Table
          Df Sum Sq Mean Sq F value    Pr(>F)
x1         1 3678.4  3678.4 34.7439 1.124e-05 ***
x2         1  402.8   402.8  3.8044   0.06603 .
x3         1   52.4    52.4  0.4951   0.49021
Residuals 19 2011.6   105.9
```

# §1.24. PS example: classroom questions I

From the fitted results, obtain the following:

(i) The fitted regression function;

(ii) The ordinary ANOVA table;

(iii) The proportion of the variation in the score of satisfaction explained by the three covariates;

(iv) The multiple correlation coefficient.

Solution:

(i) The fitted regression function is given by

$$\hat{Y} = 162.8759 - 1.2103X_1 - 0.6659X_2 - 8.613X_3.$$

(ii) From the sequential ANOVA table, it is obtained that

$$SSR = 3678.4 + 402.8 + 52.4 = 4133.6, SSE = 2011.6,$$
$$SST = 4133.6 + 2011.6 = 6145.2.$$

# §1.25. PS example: classroom questions I (cont.)

(ii) (cont.) Alternatively, since $F = MSR/MSE$,

$$MSR = F*MSE = 13.01*10.29^2 = 1377.552, \; SSR = 3*1377.552 = 4132.656.$$

(The difference is caused by rounding error.)

The ordinary ANOVA table is then:

| Source | DF | SS | MS | F-statistic |
|---|---|---|---|---|
| Regression | 3 | 4133.6 | 1377.867 | 13.01 |
| Error | 19 | 2011.6 | 105.9 | |
| Total | 22 | 6145.2 | | |

(iii) The proportion is given by $R^2 = 0.6727$.

(iv) The multiple correlation coefficient is computed as

$$\mathrm{mcorr}(\boldsymbol{y}, X) = \sqrt{R^2} = \sqrt{0.6727} = 0.8202.$$

# §2.1. The distributional properties of the sums of squares

(i) A lemma: if two random vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ are independent, then $f(\boldsymbol{x})$ and $g(\boldsymbol{y})$ are independent for any $f$ and $g$.

(ii) SSR and SSE are independent, since

$$\begin{aligned}\mathrm{Cov}(\boldsymbol{y}_R, \boldsymbol{e}) &= \mathrm{Cov}(H_R\boldsymbol{y}, H_E\boldsymbol{y}) = H_R\mathrm{Var}(\boldsymbol{y})H_E^\top \\ &= \sigma^2[X(X^\top X)^{-1}X^\top - \frac{\mathbf{1}\mathbf{1}^\top}{n}](I - X(X^\top X)^{-1}X^\top) = 0.\end{aligned}$$

(ii) $\mathrm{SSE}/\sigma^2 = (n-p-1)\hat{\sigma}^2/\sigma^2 \sim \chi^2_{n-p-1}$. It follows from the distribution of quadratic forms of standard normal vector, since

$$\begin{aligned}&\frac{1}{\sigma^2}\boldsymbol{y}^\top[I - X(X^\top X)^{-1}X^\top]\boldsymbol{y} \\ =\ &\frac{1}{\sigma^2}(X\boldsymbol{\beta} + \boldsymbol{\epsilon})^\top[I - X(X^\top X)^{-1}X^\top](X\boldsymbol{\beta} + \boldsymbol{\epsilon}) \\ =\ &\frac{1}{\sigma^2}\boldsymbol{\epsilon}^\top[I - X(X^\top X)^{-1}X^\top]\boldsymbol{\epsilon} \\ =\ &\tilde{\boldsymbol{\epsilon}}^\top[I - X(X^\top X)^{-1}X^\top]\tilde{\boldsymbol{\epsilon}}, \quad \tilde{\boldsymbol{\epsilon}} = \boldsymbol{\epsilon}/\sigma \sim N(0, I).\end{aligned}$$

(iii) Under the hypothesis that $\beta_1 = \cdots = \beta_p = 0$, $\mathrm{SSR}/\sigma^2 \sim \chi^2_p$.

# §2.2. Distributional properties of LSE (1)

(i) The mean and variance of $\hat{\boldsymbol{\beta}}$:

$$\begin{aligned} E\hat{\boldsymbol{\beta}} &= E(X^\top X)^{-1}X^\top \boldsymbol{y} = E(X^\top X)^{-1}X^\top X\boldsymbol{\beta} = \boldsymbol{\beta}. \\ \mathrm{Var}(\hat{\boldsymbol{\beta}}) &= \mathrm{Var}((X^\top X)^{-1}X^\top \boldsymbol{y}) = (X^\top X)^{-1}X^\top \mathrm{Var}(\boldsymbol{y})X(X^\top X)^{-1} \\ &= \sigma^2(X^\top X)^{-1}X^\top X(X^\top X)^{-1} = \sigma^2(X^\top X)^{-1}. \end{aligned}$$

(ii) The distribution of $\hat{\boldsymbol{\beta}}$: $\hat{\boldsymbol{\beta}}$ is a linear transformation of $\boldsymbol{y}$ which is multivariate normal, hence

$$\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2(X^\top X)^{-1}).$$

(iii) $\hat{\sigma}^2$ and $\hat{\boldsymbol{\beta}}$ are independent, since

$$\begin{aligned} \mathrm{Cov}(\hat{\boldsymbol{\beta}}, \boldsymbol{e}) &= \mathrm{Cov}((X^\top X)^{-1}X^\top \boldsymbol{y}, (I - X(X^\top X)^{-1}X^\top)\boldsymbol{y}) \\ &= \sigma^2(X^\top X)^{-1}X^\top(I - X(X^\top X)^{-1}X^\top) = 0. \end{aligned}$$

# §2.3. Distributional properties of LSE (2)

(iv) Let $\hat{\Sigma} = \hat{\sigma}^2(X^\top X)^{-1}$. Let $\hat{\sigma}_j^2$ denote the $j$th diagonal elements of $\hat{\Sigma}$ (the estimated variance of $\hat{\beta}_j$). Note that $\hat{\sigma}_j^2 = c_{jj}\hat{\sigma}^2$, where $c_{jj}$ is the $j$th diagonal element of $(X^\top X)^{-1}$.

- $(\hat{\beta}_j - \beta_j)/\hat{\sigma}_j \sim t_{n-p-1}$, since

$$(\hat{\beta}_j - \beta_j)/\hat{\sigma}_j = \frac{(\hat{\beta}_j - \beta_j)/\sigma_j}{\hat{\sigma}_j/\sigma_j} = \frac{(\hat{\beta}_j - \beta_j)/\sigma_j}{\sqrt{\hat{\sigma}^2/\sigma^2}},$$

  where $(n-p-1)\hat{\sigma}^2/\sigma^2 \sim t_{n-p-1}$, $\hat{\beta}_j$ and $\hat{\sigma}^2$ are independent.

- For any constant vector $\boldsymbol{c}$, $\dfrac{\boldsymbol{c}^\top(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})}{\sqrt{\boldsymbol{c}^\top\hat{\Sigma}\boldsymbol{c}}} \sim t_{n-p-1}$.

(v) $MSR/MSE \sim F_{p,n-p-1}$ under the hypothesis that $(\beta_1, \ldots, \beta_p) = 0$. It is shown previously that $\mathrm{SSE}/\sigma^2 \sim \chi^2_{n-p-1}$, $\mathrm{SSR}/\sigma^2 \sim \chi^2_p$ under the above hypothesis, and, in addition, SSE and SSR are independent.

$$\frac{MSR}{MSE} = \frac{\mathrm{SSR}/\sigma^2/p}{\mathrm{SSE}/\sigma^2/(n-p-1)} = \frac{\chi^2_p/p}{\chi^2_{n-p-1}/(n-p-1)}$$

has the structure of a F-distribution.

# §2.4. Significance $F$-test

- The significance test answers the question: does the regression function have a significant effect on the response variable, or, equivalently, does any of the predictors have a significant effect?
- Let $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)$. The hypotheses of the significant test are
$$H_0 : \boldsymbol{\beta} = 0, \quad H_1 : \boldsymbol{\beta} \neq 0.$$
- The test statistic is given by
$$F = \frac{\text{MSR}}{\text{MSE}}.$$
Under $H_0$, $F \sim F_{p,n-p-1}$.
- For a significance level $\alpha$, reject $H_0$ if $F \geq f_{p,n-p-1}(\alpha)$ or the $p$-value $P(F_{n,n-p-1} \geq F) < \alpha$; otherwise, do not reject $H_0$.

# §2.5. Wald statistic and its relation with $F$-statistic

- Let $\boldsymbol{\theta}$ be a vector of parameters, $\hat{\boldsymbol{\theta}}$ its estimator, and $\hat{\Sigma}_\theta$ the estimated variance matrix of $\hat{\boldsymbol{\theta}}$. The hypotheses $H_0 : \boldsymbol{\theta} = 0$ v.s. $H_1 : \boldsymbol{\theta} \neq 0$ can be tested by using the Wald statistic defined below:
$$\hat{\boldsymbol{\theta}}^\top \hat{\Sigma}_\theta^{-1} \hat{\boldsymbol{\theta}}.$$
- The Wald statistic for the significance test is given by
$$W = \hat{\boldsymbol{\beta}}^\top \hat{\Sigma}_\beta^{-1} \hat{\boldsymbol{\beta}},$$
where $\hat{\boldsymbol{\beta}}$ is the estimate of $\boldsymbol{\beta}$ and $\hat{\Sigma}_\beta$ is the estimated variance matrix of $\hat{\boldsymbol{\beta}}$.
- In the context of multiple LRM, the $F$-statistic and Wald statistic has the relation: $F = W/p$, where $p$ is the dimension of $\boldsymbol{\beta}$.

# §2.6. PS example: $F$ and Wald statistic

- The value of the $F$-statistic is given in the fitted results as $F = 13.01$ with d.f. 3 and 19, and $p$-value `7.482e-5`.
- The Wald statistic is computed by R as follows:

```
bb = PS.fit$coef; VV = vcov(PS.fit)
b = bb[-1]; V = VV[-1,-1]
W =t(b)%*%solve(V)%*%b; F= W/3
```

bb and `VV` are:

```
bb:   162.8758987  -1.2103182   -0.6659056  -8.6130315
VV:   664.3842308  0.10170959 -15.99369995  64.6405236
        0.1017096  0.09087306  -0.03344659  -0.8723306
      -15.9936999 -0.03344659   0.67403600  -7.3664602
       64.6405236 -0.87233063  -7.36646020 149.8482325
```

$W = 39.04337, \ F = 39.04337/3 = 13.01446.$

# §2.7. Individual $t$-test

- The individual $t$-test answers the question: after taking into account the effects of the other variables in the model, does a particular predictor still have a significant effect?
- The hypotheses:
$$H_0 : \beta_j = 0 \text{ versus } H_1 : \beta_j \neq 0.$$
- The test statistic is given by
$$T = \hat{\beta}_j / s.d(\hat{\beta}_j),$$
where $s.d(\hat{\beta}_j) = \hat{\sigma}_j$ is the estimated standard deviation of $\hat{\beta}_j$. Under $H_0$, $T \sim t_{n-p-1}$.
- For a significance level $\alpha$, reject $H_0$ if $|T| \geq t_{n-p-1}(\alpha/2)$ or the $p$-value $2P(t_{n-p-1} \geq |T|) < \alpha$; otherwise, do not reject $H_0$.
- One-sided tests are carried out in the same way as in the case of simple LRM.

# §2.8. Testing general linear hypothesis

- The general linear hypothesis is of the form
$$H_0 : \sum_{j=0}^{p} c_j \beta_j = 0.$$
- For testing the linear hypothesis, the test statistic is given by
$$T = \frac{\boldsymbol{c}^\top \hat{\boldsymbol{\beta}}}{\sqrt{\boldsymbol{c}^\top \hat{\Sigma} \boldsymbol{c}}}.$$
Under $H_0$, $t \sim t_{n-p-1}$.
- The formula for the statistic can be simplified when only a few components of $\boldsymbol{c}$ are nonzero. For example, if $\boldsymbol{c} = (0, c_1, c_2, 0, \ldots, 0)^\top$, the variance $\boldsymbol{c}^\top \hat{\Sigma} \boldsymbol{c}$ becomes $c_1^2 \mathrm{Var}(\hat{\beta}_1) + c_2^2 \mathrm{Var}(\hat{\beta}_2) + 2c_1 c_2 \mathrm{Cov}(\hat{\beta}_1, \hat{\beta}_2)$.
- For a significance level $\alpha$, reject $H_0$ if $|T| \geq t_{n-p-1}(\alpha/2)$ or the $p$-value $2P(t_{n-p-1} \geq |T|) < \alpha$; otherwise, do not reject $H_0$.

# §2.9. Confidence interval and confidence bound

- A $100(1-\alpha)\%$ confidence interval for $\beta_j$ is given by

$$[\hat{\beta}_j - \hat{\sigma}_j t_{n-p-1}(\alpha/2),\ \hat{\beta}_j + \hat{\sigma}_j t_{n-p-1}(\alpha/2)].$$

- A $100(1-\alpha)\%$ confidence interval for $\boldsymbol{c}^\top \boldsymbol{\beta}$ is given by

$$[\boldsymbol{c}^\top \hat{\boldsymbol{\beta}} - \sqrt{\boldsymbol{c}^\top \hat{\Sigma} \boldsymbol{c}} * t_{n-p-1}(\alpha/2),\ \boldsymbol{c}^\top \hat{\boldsymbol{\beta}} - \sqrt{\boldsymbol{c}^\top \hat{\Sigma} \boldsymbol{c}} * t_{n-p-1}(\alpha/2)].$$

- The $100(1-\alpha)\%$ confidence bounds for $\beta_j$ are given by

$$\begin{aligned} \text{Upper bound}: &\quad \beta_j \le \hat{\beta}_j + \hat{\sigma}_j t_{n-p-1}(\alpha), \\ \text{Lower bound}: &\quad \beta_j \ge \hat{\beta}_j - \hat{\sigma}_j t_{n-p-1}(\alpha). \end{aligned}$$

# §2.10. Predictions

- Given a new observation $\boldsymbol{x}_0 = (1, x_{01}, \ldots, x_{0p})^\top$, the predicted value for both $Ey_0$ and $y_0$ is $\hat{y}_0 = \boldsymbol{x}_0^\top \hat{\boldsymbol{\beta}}$.
- The estimated variance of the fitted value is given by

$$\hat{\sigma}_F^2(\hat{y}_0) = \boldsymbol{x}_0^\top \mathrm{Var}(\hat{\boldsymbol{\beta}})\boldsymbol{x}_0 = \boldsymbol{x}_0^\top \hat{\Sigma}\boldsymbol{x}_0.$$

- The estimated prediction error variance is given by

$$\hat{\sigma}_P^2(y_0) = \boldsymbol{x}_0^\top \hat{\Sigma}\boldsymbol{x}_0 + \hat{\sigma}^2 = \hat{\sigma}^2 + \hat{\sigma}_F^2(\hat{y}_0).$$

- The confidence interval for $Ey_0$ is given by

$$\hat{y}_0 \pm \hat{\sigma}_F(\hat{y}_0) t_{n-p-1}(\alpha/2).$$

- The prediction interval for $y_0$ is given by

$$\hat{y}_0 \pm \hat{\sigma}_P(y_0) t_{n-p-1}(\alpha/2).$$

# §2.11. PS example: classroom questions II

The fitted results are reproduced below:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 162.8759    25.7757   6.319 4.59e-06 ***
x1           -1.2103     0.3015  -4.015  0.00074 ***
x2           -0.6659     0.8210  -0.811  0.42736
x3           -8.6130    12.2413  -0.704  0.49021
Residual standard error: 10.29 on 19 degrees of freedom
F-statistic: 13.01 on 3 and 19 DF,  p-value: 7.482e-05
Estimated varinace-covariance matrix:
           (Intercept)      x1       x2       x3
(Intercept)    664.3842  0.1017 -15.9937  64.6405
x1               0.1017  0.0909  -0.0334  -0.8723
x2             -15.9937 -0.0334   0.6740  -7.3665
x3              64.6405 -0.8723  -7.3665 149.8482
```

(i) Conduct the significant test to conclude whether the regression function has a significant effect on the response.

(ii) Construct a 95% C.I. for $\beta_1$.

# §2.12. PS example: classroom questions II (cont.)

(iii) It is reasonable to suspect that the satisfaction of patients could decrease as their age, severity of illness and anxiety increase. Formulate appropriate tests to confirm or disconfirm the suspect.

(iv) Construct an upper 95% confidence bound for $\beta_1$.

(v) Construct an 95% prediction interval for a patient with `(x1,x2,x3) = (50,51,2.3)`

Solution:

(i) The F-statistic for the significant test has value 13.01 and p-value `7.482e-05`. With the small p-value, it can be concluded that the regression function has a significant effect on the response.

(ii) The upper $\alpha/2$-quantile is obtained by `qt(0.975, 19)` as 2.093 for $\alpha = 0.05$. The 95% C.I. for $\beta_1$ is computed as

$$-1.2103 \pm 0.3015 * 2.093 = [-1.8413,\ -0.5793].$$

# §2.13. PS example: classroom questions II (cont.)

(ii) (cont.) Note: Confidence intervals for other parameters can be computed in the same way. The R function `confint` together with the fitted object can be used to compute confidence intervals for all parameters as follows

```
> confint(PS.fit, level = 0.95)
(Intercept) 108.926839 216.8249581
x1           -1.841264  -0.5793727
x2           -2.384272   1.0524608
x3          -34.234265  17.0082018
```

(iii) The hypotheses for the tests are as follows: ( Why?)

$$H_{j0} : \beta_j \geq 0, \quad H_{j1} : \beta_j < 0, \quad j = 1, 2, 3.$$

The test statistic given in the fitted results are compared with `-qt(0.95, 19) = -1.7291`. Only $H_{10}$ with test statistic $-4.015$ is rejected, the others are not rejected.

The $p$-values of the tests can be obtained from the fitted results. The $p$-value of the one-sided test is half that of the two-sided test. The $p$-value of the two-sided test in the summary table in the column `Pr(>|t|)`. E.g., for $H_{10}$, the p-value is $0.00074/2 = 0.00037$.

# §2.14. PS example: classroom questions II (cont.)

(iv) The upper 95% confidence bound for $\beta_1$ is

$$\beta_1 \leq -1.21 + 0.3015 * 1.7291 = -0.6887.$$

Bounds for other parameters are computed in the same way.

(v) Let $\boldsymbol{x}_0 = (1, 50, 51, 2.3)$. The prediction interval is computed by

```
b = PS.fit$coef; V = vcov(PS.fit); sigma=10.29
x0=c(1,50,51,2.3); y0=t(x0)%*%b
sP = sqrt(t(x0)%*%V%*%x0 +sigma^2)
t = qt(0.975,19)
cI = c(y0-sP*t, y0+sP*t)
```

which produces the interval [25.65226, 71.52541].

The prediction can also be computed using function `predict` as follows:

```
> xnew =list(x1=50,x2=51,x3=2.3)
> predict(PS.fit,newdata=xnew,interval="prediction",level=.95)
          fit      lwr      upr
    1 48.58883 25.65333 71.52433
```

(Differences are due to rounding error.)